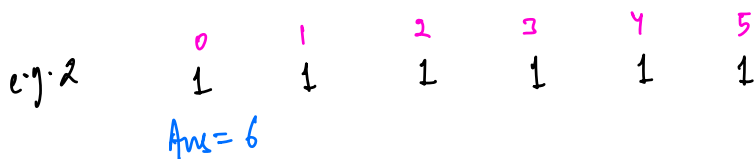
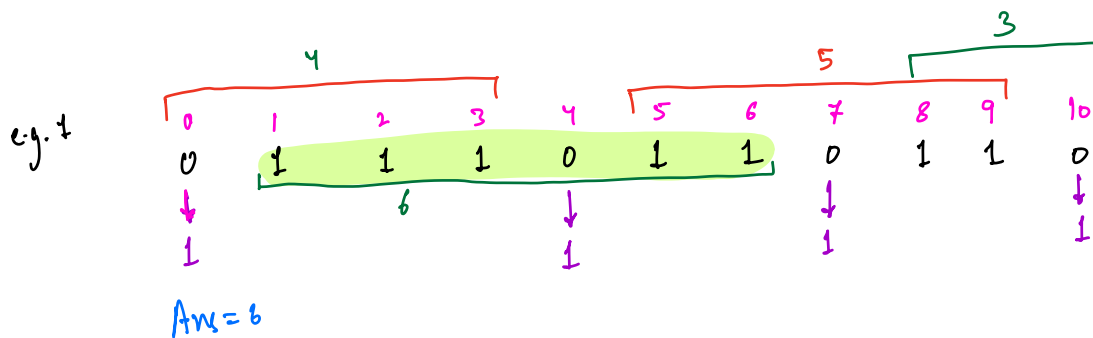


Today's content :-

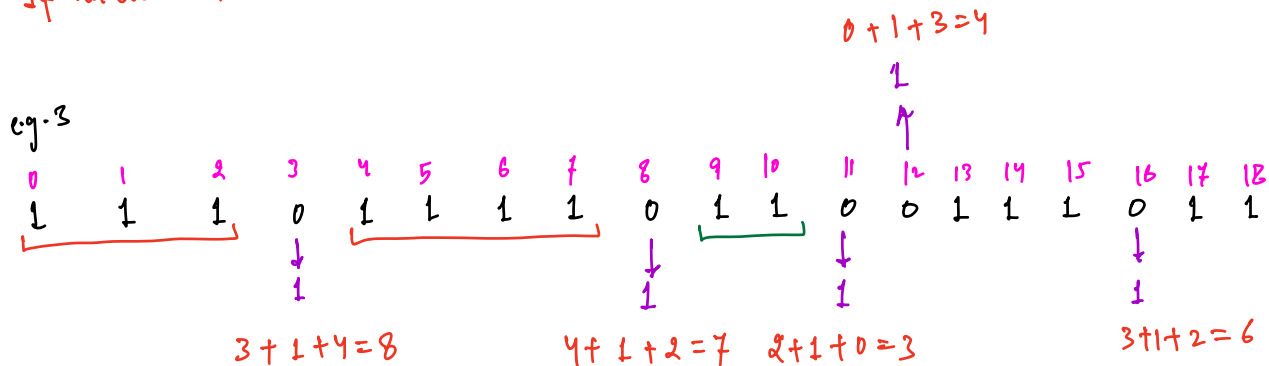
- 1) Max consecutive 1's (replace 0)
- 2) Max consecutive 1's (swap 0)
- 3) Josephus

Ques: Given a binary array $\rightarrow \{0, 1\}$ $\rightarrow$ max upto 1

Microsoft
LinkedIn
Amazon We are allowed to replace almost one 0 with a 1.
Return length of max. consecutive 1s.



If all elements are 1's, ans = N.



Ans = 8

Iterate on arr[]:-

if (arr[i] == 0) {

left = no. of consecutive 1's on left side of i
 right = no. of consecutive 1's on right side of i
 ans = max(ans, left + right + 1);
 }

// Pseudo Code

```
int ans = 0;
int n = arr.length;
int c = 0;
for (i = 0; i < N; i++) { c = c + arr[i]; }
if (c == N) return N;
```

```
for (int i = 0; i < N; i++) {
    if (arr[i] == 0) {
        // l = no. of cons. 1's on left of i
        for (j = i - 1; j >= 0; j--) {
            if (arr[j] == 1) l++;
            else break;
        }
        // r = no. of cons. 1's on right of i
        for (j = i + 1; j < N; j++) {
            if (arr[j] == 1) r++;
            else break;
        }
        ans = max(ans, l + r + 1);
    }
}
return ans;
```

arr[1, N]

```
for (l = 1; l <= 9; l++) {
    for (i = 0; i < N; i++) {
        print(arr[i]);
    }
}
```

Edge case:-

1) arr[4] = 1 1 1 1

Correct ans = 4

our code returns 0;

2) arr[3] = 0 0 0

Diagram illustrating the calculation of the maximum number of consecutive 1's for each index i:

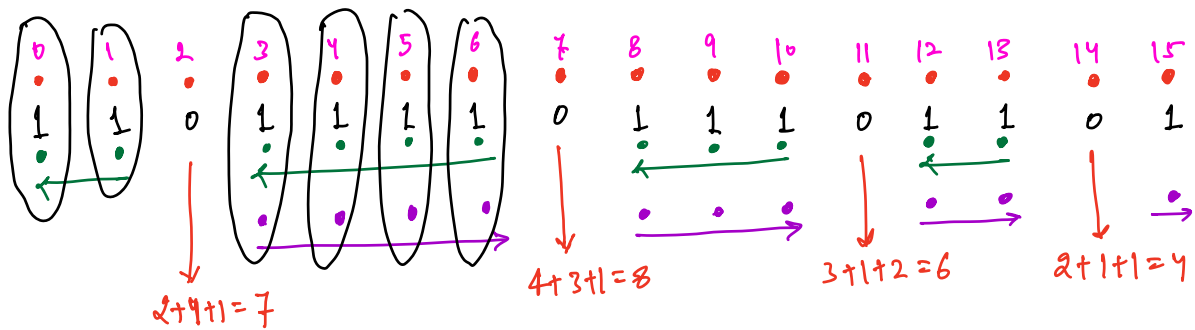
- For i = 0: l = 0, r = 0, l + r + 1 = 1
- For i = 1: l = 0, r = 0, l + r + 1 = 1
- For i = 2: l = 0, r = 0, l + r + 1 = 1

ans = 1

T.C = ~~$O(N^2)$~~ $O(N)$
S.C = $O(1)$

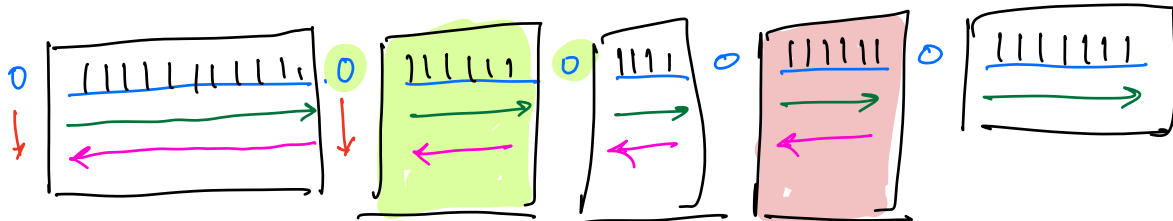
iterations = $4 * N$

eg.



$$T.C = 3 * N = O(N)$$

Another generalised example.



Note:- Whenever there are nested for loops with a break statements, be careful about T.C. T.C. in such cases can be $O(N)$

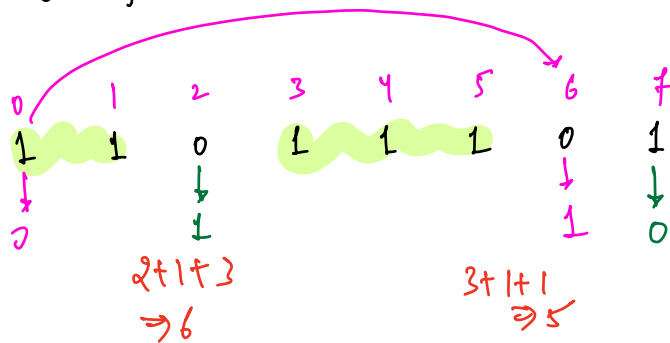
Ques 2 Max consecutive 1's

Given a binary array $\rightarrow \{0, 1\}$

We are allowed to swap a single 0 with a 1.

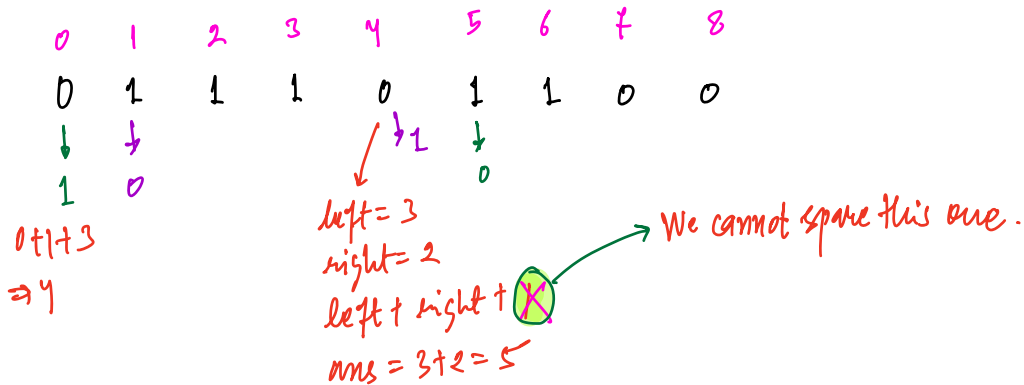
Return length of max. consecutive 1's.

e.g.



Ans = 6

e.g.



→ Since, we cannot spare an extra 1 for this, count = left + right

→ If we are able to spare an extra 1, count = left + right + 1

→ How to know, if you can spare an extra 1 or not

if (left) = total no. of 1's → we cannot spare.

// Pseudo Code

int ans = 0

int n = arr.length;

int c = 0;

for (i = 0; i < N; i++) { c = c + arr[i]; }

if (c == N) return N;

for (int i = 0; i < N; i++) {

if (arr[i] == 0) {

// l = no. of cons. 1's on left of i

for (j = i - 1; j >= 0; j--) {

if (arr[j] == 1) l++;

else break;

// r = no. of cons. 1's on right of i

for (j = i + 1; j < N; j++) {

if (arr[j] == 1) r++;

else break;

// cannot spare

if (l + r == c) { ans = max(ans, l + r); }

else { ans = max(ans, l + r + 1); }

}

}

return ans;

X, y

Break → 10:45 PM

Ques 3 Given N , how many perfect square $[1-N]$

$$N=25 \rightarrow [1, 4, 9, 16, 25]$$

$1^2 \quad 2^2 \quad 3^2 \quad 4^2 \quad 5^2$

$$N=50 \rightarrow [1, 4, 9, 16, 25, 36, 49]$$

$1^2 \quad 2^2 \quad 3^2 \quad 4^2 \quad 5^2 \quad 6^2 \quad 7^2$

$$\text{ans} = \text{sqrt}(N) \rightarrow \text{integer part}$$

$$i^2 \leq N$$

$$i \leq \sqrt{N}$$

$$\max(i) = \sqrt{N}$$

Ques 1 Puzzle → Josephus Punishment

Initially N people in a circle.

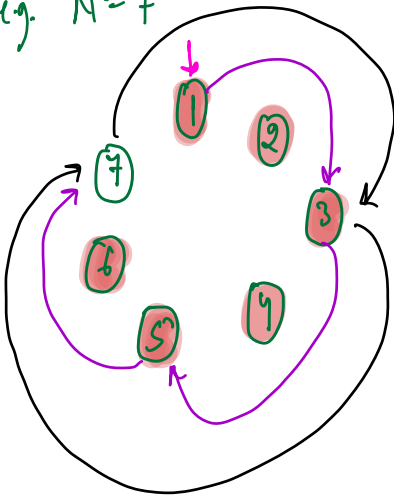
Person 1 has a knife, and kills the clockwise adjacent person.

Also, this person passes the knife to the next adjacent clockwise person.

Repeat this, until only a single person stands.

Return this single person.

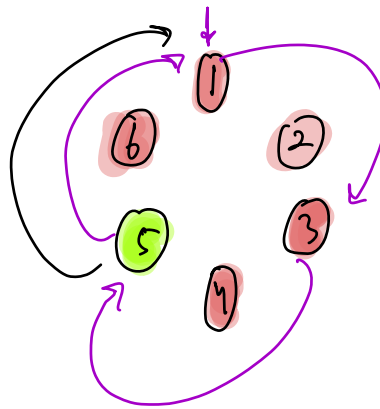
e.g. $N=7$



When, $N=7 \rightarrow \text{winner}=7$

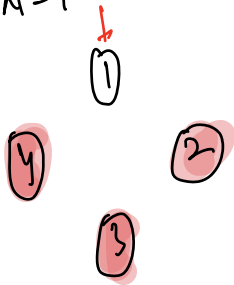
When, $N=6 \rightarrow \text{winner}=5$

e.g. $N=6$



If n is odd $\rightarrow N$
If n is even $\rightarrow N-1$ } $\rightarrow$ X 100% wrong.

e.g. $N=4$

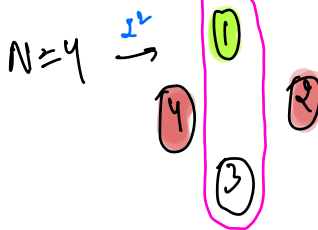


Winner = 1

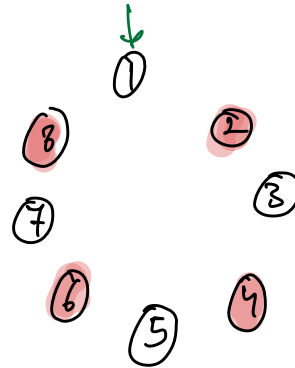
$N=1 \xrightarrow{2^0} 1$

$N=2 \xrightarrow{2^1} 1$

2

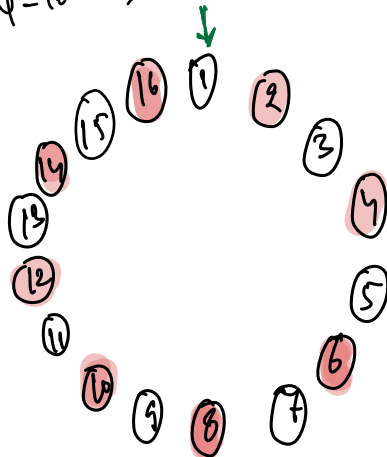


$N=8 \xrightarrow{2^3}$



Winner = 1

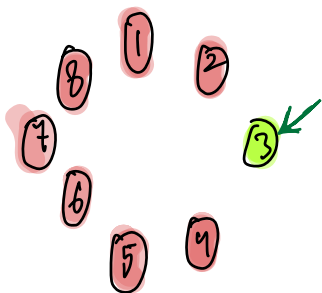
$N=16 \xrightarrow{2^4}$



Ans = 1

Observation: If N is a power of 2 and we start at 1, then 1 is winner.

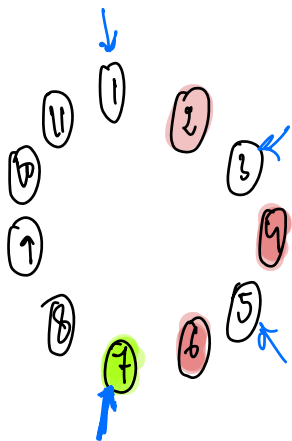
e.g.



First observation

→ If N is a power of 2 and we start at x , then x is the winner

$N=11$



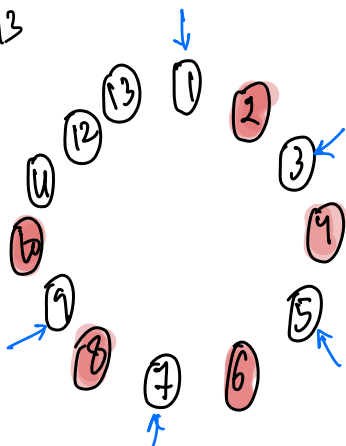
at this instant 8 people are alive.

Now,

$N=11 \xrightarrow{3 \text{ kills}} 8$
(nearest power of 2 < 11)

$1 \rightarrow 3 \rightarrow 5 \rightarrow 7$
↓
ans

$N=13$



5 kills
 $13 \rightarrow 8$

5 jumps.

$1 \rightarrow 3 \rightarrow 5 \rightarrow 7 \rightarrow 9 \rightarrow 11$

$\Rightarrow 2 * 5 + 1 = 11$

e.g. $N=9 \xrightarrow{1 \text{ kill}} 8$
1 jump $\rightarrow 2 * 1 + 1 = 3$

e.g. $N=100 \xrightarrow{\quad} 64$
 $\downarrow$
36 kills.

$$2 * 36 + 1 \Rightarrow 73$$

e.g. $N=16 \xrightarrow{\text{0 kills}} 16$

No jumps.

$$2 * 0 + 1 \Rightarrow 1$$

Pseudo Code:-

```
alive(N) {  
  |  $p = \text{Nearest power of } 2 \leq N$  → TODO  
  |  $\text{return } 2 * (N - p) + 1$   
  |  
}
```

————— X ——— DONE ——— X —————

Interview Problems 2.